

An Unconventional View on Beta-Reduction in Namefree Lambda-Calculus

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Dedicated to Stefano Berardi on the occasion of his 64th birthday

Overview

A function call in untyped λ -calculus corresponds to a β -reduction step, whose most common form is: $(\lambda x.T)(V) \rightarrow_\beta T[V/x]$.

The substitution $T[V/x]$ replaces with V **all** free instances of x in T , while avoiding captures through α -conversion, i.e., renaming.

From the computational standpoint, avoiding captures is a time consuming operation that reduction machines strive to avert.

Replacing named variable instances with de Bruijn's depth indexes (**nameless** binding) speeds up this operation without nullifying it.

The depth index of a variable instance x bound in T is the number of λ 's of T in whose scope x occurs free. Thus, it's a positive integer.

The nameless substitution $T[V/1]$ may update the depth indexes of the free variables in the copies of V that it inserts in T .

This operation, known as lifting, amounts to applying an **update function** τ_h (where $h \geq 0$) to the depth indexes occurring in V .

Our proposal

We recall that $\tau_h(n) \stackrel{\text{def}}{=} n + h$ appears first in early de Bruijn's works, where the theory of update functions finds its foundation.

A class of systems originating from these works and from $\lambda\sigma$ **delay** the application of update functions by storing them in their syntax.

The main reason for so doing is allowing a computation either to delay updates at will or to apply them whenever is the case.

The system we are presenting here today belongs to this class.

We wish to introduce a β -reduction step $\rightarrow_{\text{ef}}$ that is **focused** and **expanding**. We explain these terms by means of an example.

$$(\lambda x.x(x))(V) = (\lambda.1(\mathbf{1}))(V) \rightarrow_{\text{ef}} (\lambda.1(\mathbf{1}.V))(V)$$

The step attaches a copy of V after the leaf $\mathbf{1}$, which becomes an inner node and records that we still need to update V with $\tau_{\mathbf{1}}$.

Focused means we replace exactly one instance of x .

Expanding means the redex is a subtree of the reduct.

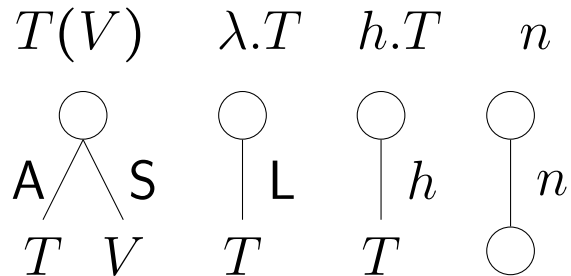
Our requirements

“Expanding” requires two features. The construct $h.T$ representing the delayed lift of the term T by τ_h and β -reduction at a distance.

This β -reduction appears first in Nederpelt’s works and for instance allows the step: $(\lambda x_1.\lambda x_2.T)(V_1)(V_2) \rightarrow_{\beta d} (\lambda x_1.T[V_2/x_2])(V_1)$.

“Focused” requires to single out the instance of the variable we want to replace. To this aim we use a **path in the redex a.s.t.**

The a.s.t.’s we are proposing have no labels on nodes, just on edges.



We denote a path by the labels of its edges listed from the root (left end) to the leaf (right end). E.g., the path to the leaf 1 is: **A L S 1**.

Once paths are introduced, representing a term as the **set of all paths in its a.s.t.** allows to operate easily on a single path.

Our main definitions

In this setting $T[W/V]$ replaces the set of paths V with W in T and $pT \stackrel{\text{def}}{=} \{pq \mid q \in T\}$ for p and q arbitrary segments of paths.

Definition (subtree). Let $\text{tree}_T(p) \stackrel{\text{def}}{=} \{q \mid pq \in T\}$.

Balanced segments are a key notion in reduction at a distance.

Definition (balanced segment). ϵ (the empty path segment) is balanced, if b and c are balanced so are: AbL , hb , bh , bc .

When $pLq(n+1) \in T$ and $n \in \mathbb{N}$, $n+1$ **refers to the displayed** L when $\|q\| = n$, where $\|q\|$ counts the L 's in q being aware of lifts.

Definition (ef-reduction). If $r = pAbLq(n+1) \in T$, b is balanced and $\|q\| = n \in \mathbb{N}$ then $T \xrightarrow{r}_{\text{ef}} T[r \text{tree}_T(pS)/r]$.

Key observation 1. The delayed lifts in the segment b are never applied to any variable occurrence and thus we must rule them out.

Key observation 2. The introduced delayed lift $n+1$ does not denote τ_{n+1} but $\tau_{|b|+n+1}$, where $|b|$ counts the labels L in b .

Concluding remarks

Observations 1 and 2 imply that the construct $h.T$ denotes the application of some $\tau_{h'}$ to T , where h' depends on context of $h.T$.

An **algorithm** exists to check if a path $r \in T$ has the form $p \mathbf{A} b \mathbf{L} q$ ($n + 1$) in which b and q meet the requirements for ef-reduction.

We can give a variant of the ef-reduction grounded in the assumption that $h.T$ denotes the context independent application of τ_h to T .

Under the assumptions of ef-reduction, $T \xrightarrow{r}_{\text{ef}'} T[r' \text{ tree}_T(p\mathbf{S})/r]$ where $r' \stackrel{\text{def}}{=} p \mathbf{A} \underline{b} \mathbf{L} q$ ($|b| + n + 1$) and $\underline{b}$ sets to 0 the labels h of b .

To **apply** the delayed lifts in a path $p n \in T$ we remove them from p and we apply to n an update function (unfold) depending on p .

We can express other reductions in this system as well. For instance variants of non-expanding focused β -reduction and ζ -reduction.

We are verifying that all these reductions have desired properties, e.g., **confluence**. Such results will appear in a forthcoming article.

Thank you

Appendix

When $h.T$ denotes the delayed application of τ_h to T (i.e., this construct has a context-free semantics), we can set the next definitions.

Definition (depth). $\|p\| \in \mathbb{N}$ is given by: $\|\epsilon\| \stackrel{\text{def}}{=} 0$, $\|pA\| \stackrel{\text{def}}{=} \|pS\| \stackrel{\text{def}}{=} \|p\|$, $\|pL\| \stackrel{\text{def}}{=} \|p\| + 1$, if $\|p\| \geq h$ then $\|ph\| \stackrel{\text{def}}{=} \|p\| - h$.

The inductive definition of the unfold update function $f \circ p$ requires a parameter f whose initial value is the **identity function**.

Definition (push). The push $\mathbin{\circlearrowleft} f$ of a function f is a well-known function we define by: $(\mathbin{\circlearrowleft} f)(1) \stackrel{\text{def}}{=} 1$, $(\mathbin{\circlearrowleft} f)(n+1) \stackrel{\text{def}}{=} (\mathbin{\circlearrowleft} f)(n) + 1$.

Definition (unfold). The function $f \circ p$ is given by: $f \circ \epsilon \stackrel{\text{def}}{=} f$, $f \circ pA \stackrel{\text{def}}{=} f \circ pS \stackrel{\text{def}}{=} f \circ p$, $f \circ pL \stackrel{\text{def}}{=} (\mathbin{\circlearrowleft} f) \circ p$, $f \circ ph \stackrel{\text{def}}{=} (f \circ p) \circ \tau_h$.

Applying an update function f to a path (lifting) requires the well-known function that we define by: $(f \mathbin{\circlearrowright} h)(n) \stackrel{\text{def}}{=} f(h+n) - f(h)$.

Definition (lift). The lift $\uparrow^f p$ is defined by: $\uparrow^f \epsilon \stackrel{\text{def}}{=} \epsilon$, $\uparrow^f Ap \stackrel{\text{def}}{=} A \uparrow^f p$, $\uparrow^f Sp \stackrel{\text{def}}{=} S \uparrow^f p$, $\uparrow^f Lp \stackrel{\text{def}}{=} L \uparrow^{\mathbin{\circlearrowleft} f} p$, $\uparrow^f hp \stackrel{\text{def}}{=} f(h) \uparrow^{f \mathbin{\circlearrowright} h} p$.